Series GEFH1/C

SET ~ 1

प्रश्न-पत्र कोड
Q.P. Code **55/C/1**

रोल नं.

Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book. *

भौतिक विज्ञान (सैद्धान्तिक)
PHYSICS (Theory)

निर्धारित समय : 3 घण्टे | *अधिकतम अंक : 70*

Time allowed : 3 hours | *Maximum Marks : 70*

नोट / NOTE :

(i) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **27** हैं ।
Please check that this question paper contains **27** printed pages.

(ii) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(iii) कृपया जाँच कर लें कि इस प्रश्न-पत्र में **35** प्रश्न हैं ।
Please check that this question paper contains **35** questions.

(iv) **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
Please write down the serial number of the question in the answer-book before attempting it.

(v) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **35** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*

(iii) *In **Section A** – Questions no. **1** to **18** are Multiple Choice (MCQ) type questions, carrying **1** mark each.*

(iv) *In **Section B** – Questions no. **19** to **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C** – Questions no. **26** to **30** are Short Answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D** – Questions no. **31** to **33** are Long Answer (LA) type questions carrying **5** marks each.*

(vii) *In **Section E** – Questions no. **34** and **35** are case-based questions carrying **4** marks each.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 3 questions in Section D and 2 questions in Section E.*

(ix) *Use of calculators is **not** allowed.*

Use the following values of physical constants, if required :

$c = 3 \times 10^{8}$ m/s

$h = 6{\cdot}63 \times 10^{-34}$ Js

$e = 1{\cdot}6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

$\varepsilon_0 = 8{\cdot}854 \times 10^{-12}$ C^2 N^{-1} m^{-2}

$\dfrac{1}{4\pi\varepsilon_0} = 9 \times 10^{9}$ N m^2 C^{-2}

Mass of electron (m_e) = $9{\cdot}1 \times 10^{-31}$ kg

Mass of neutron = $1{\cdot}675 \times 10^{-27}$ kg

Mass of proton = $1{\cdot}673 \times 10^{-27}$ kg

Avogadro's number = $6{\cdot}023 \times 10^{23}$ per gram mole

Boltzmann constant = $1{\cdot}38 \times 10^{-23}$ JK^{-1}

SECTION A

1. A small object lies at the bottom of a vessel filled with water (refractive index 4/3) up to a height H. When viewed from a point above the surface of water, the object appears raised by n percent of H. The value of n is :

(a) 15 (b) 20

(c) 25 (d) 33

2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)

(a) $-e(v_x B_y - v_y B_x)\hat{k}$ (b) $e(v_x B_y - v_y B_x)\hat{k}$

(c) $-e(v_x B_y + v_y B_x)\hat{k}$ (d) $e(v_x B_y + v_y B_x)\hat{k}$

3. A bar magnet is cut into two equal halves parallel to its magnetic axis. The physical quantity that remains unchanged is :

(a) pole strength (b) magnitude of magnetisation

(c) moment of inertia (d) magnetic moment

4. In a series LC circuit connected to an ac source, with the increase in the frequency of the source, the net reactance :

(a) increases linearly

(b) decreases linearly

(c) first increases to become maximum and then decreases to zero

(d) first decreases to become zero and then increases

5. Which of the following physical quantities remain the same for X-ray, red light and radio waves when travelling through a medium ?

(a) Wavelength (b) Speed

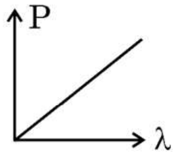
(c) Frequency (d) Momentum

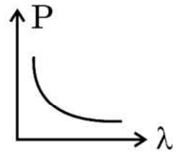
6. In Young's double-slit experiment, the intensity on the screen is I_0 at a point where path difference is λ. The intensity at the point where path difference is $\frac{\lambda}{4}$ is :

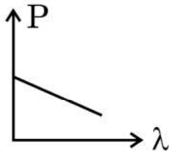
(a) $\frac{I_0}{4}$ (b) $\frac{I_0}{2}$

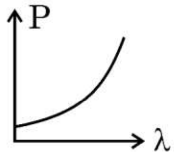
(c) I_0 (d) zero

7. Which of the following figures represents the variation of a particle's momentum with the de Broglie wavelength associated with it ?

(a) 

(b) 

(c) 

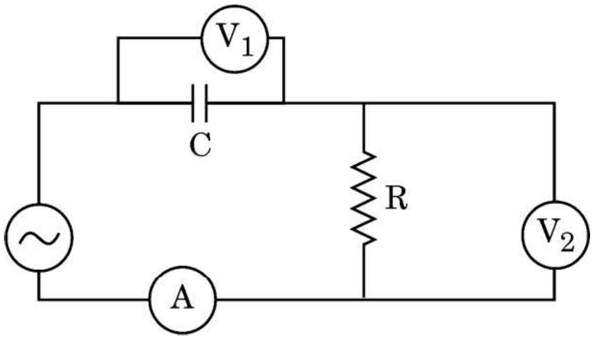
(d) 

8. In the nuclear reaction $_{7}N^{14} + {}_{2}He^{4} \longrightarrow X + {}_{1}H^{1}$, X represents :

(a) $_{7}O^{16}$ (b) $_{8}N^{17}$

(c) $_{8}O^{17}$ (d) $_{7}N^{16}$

9. The given figure shows a capacitor C and a resistor R connected in series to an ac source. V_1 and V_2 are voltmeters and A is an ammeter.



Which of the following statements is correct ?

(a) Current in the circuit lags in phase with voltage shown in V_2 .

(b) The voltage shown in V_1 is ahead in phase with voltage shown in V_2.

(c) The current in the circuit and the voltage shown in V_1 are always in phase.

(d) The voltage shown in V_1 lags behind in phase with the voltage shown in V_2 .

10. Light of frequency $1{\cdot}5\ \nu_0$ is incident on a photosensitive material of threshold frequency ν_0. If the frequency of the incident radiation is kept constant and intensity is increased, the photo current will :

(a) increase

(b) decrease

(c) not change

(d) first decrease and then become zero

11. The impact parameter for an alpha particle approaching a target nucleus is maximum when the scattering angle (θ) is :

(a) $0°$ (b) $90°$

(c) $180°$ (d) $45°$

12. Two nuclei have their mass numbers in the ratio of 1 : 27. What is the ratio of their nuclear densities ?

(a) 1 : 27 (b) 1 : 1

(c) 1 : 9 (d) 1 : 3

13. A plane wave is incident on a concave mirror of radius of curvature R. The reflected wave is a spherical wave of radius :

(a) $\frac{R}{4}$ (b) $\frac{R}{2}$

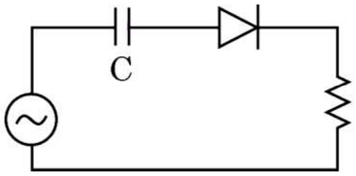
(c) R (d) 2R

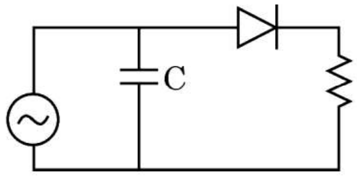
14. A planar loop is rotated in a magnetic field about an axis perpendicular to the field. The polarity of induced emf changes once in each :

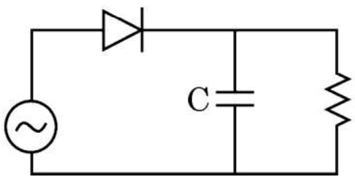
(a) 1 revolution (b) $\left(\frac{1}{2}\right)$ revolution

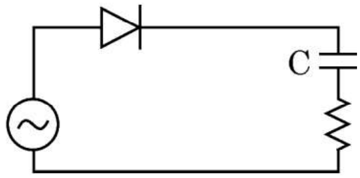
(c) $\left(\frac{1}{4}\right)$ revolution (d) $\left(\frac{3}{4}\right)$ revolution

15. In which of the following diagrams is the capacitor 'C' connected correctly to provide smooth output of a half-wave rectifier ?

(a)


(b)


(c)


(d)


*Questions number **16** to **18** are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.*

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false and Reason (R) is also false.

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

17. *Assertion (A) :* A planar loop of irregular shape carrying current is subjected to a magnetic field acting perpendicular to the plane of the loop. If the wire is flexible, the loop takes a circular shape.

Reason (R) : The force acting on each point of a current carrying loop, in a magnetic field perpendicular to its plane, is radially outward.

18. *Assertion (A) :* Silicon is preferred over germanium for making semiconductor devices.

Reason (R) : The energy gap for germanium is more than the energy gap for silicon.

SECTION B

19. (a) The electric field of an electromagnetic wave passing through vacuum is represented as $E_x = E_0 \sin (kz - \omega t)$. Identify the parameter which is related to the (i) wavelength, and (ii) the frequency of the wave in the above equation.

(b) Write two properties of a medium that determine the velocity of light in that medium. *2*

20. Two identical bars, one of a paramagnetic material and another of a diamagnetic material are kept in a uniform magnetic field. Show diagrammatically the modifications in the pattern of magnetic field in each case. How are the two materials affected by increase in temperature ? *2*

21. Light of wavelength 3500 Å is incident on two metals A and B. Which of them will yield photoelectrons, if their work functions are 4·2 eV and 1·9 eV respectively ? Make the necessary calculations to justify your answer. *2*

22. The inward and the outward electric flux through a Gaussian surface are 2ϕ and ϕ respectively.

(a) What is the net charge enclosed by the surface ?

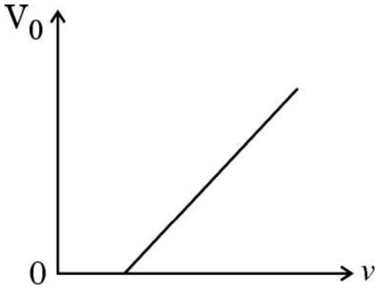
(b) If the net outward flux through the surface were zero, can it be concluded that there were no charges inside the surface ? Justify your answer. *2*

23. (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ? *2*

OR

(b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R, 2R and 3R. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer. *2*

24. The figure shows the variation of stopping potential V_0 with frequencies of incident monochromatic beam for a metal. Such a graph was first obtained by R.A. Millikan in 1916 for sodium.



Explain how, using Einstein's photoelectric equation and the graph, you can obtain the value of (i) Planck's constant, and (ii) work function of the metal, given the value of charge on electron. *2*

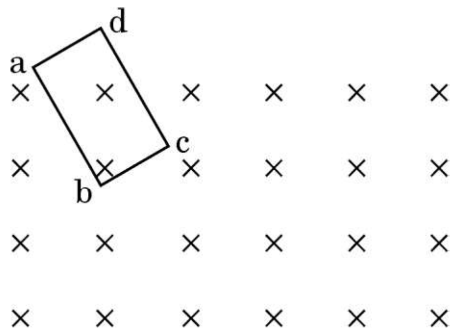
25. (a) A uniform electric field E of 500 N/C is directed along +x axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B. *2*

OR

(b) Three point charges 1 μC, –1 μC and 2 μC are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1, B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1, from B to B_1 and from C to C_1. *2*

SECTION C

26. (a) State Lenz's law. Determine the direction of the induced current when a rectangular conducting loop abcd in xy-plane is moved into a region of magnetic field which is directed along z-axis. *3*



OR

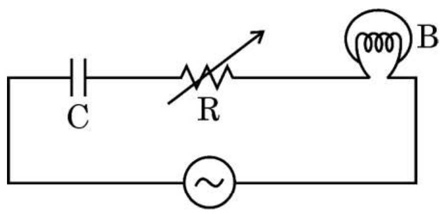
(b) Two identical circular loops, one of copper and the other of aluminium are rotated about their diameters with the same angular speed in a magnetic field directed perpendicular to their axes of rotation. Compare (i) the emf induced, and (ii) the current in the two loops. Justify your answers. *3*

27. (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?

(b) Why should the value of resistance connected in parallel to a galvanometer be low ?

(c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ? *3*

28. The figure shows a capacitor C, a variable resistor R and a bulb connected in series to the ac source of frequency (ν). The bulb glows with some brightness.



How will the glow of the bulb be affected, if the

(a) separation between the plates of the capacitor is doubled, keeping resistance R and frequency (ν) the same ?

(b) resistance R is decreased keeping the value of capacitance C and frequency (ν) same ?

(c) frequency of ac source is decreased for the same value of C and R ? Justify your answer in each case. *3*

29. (a) (i) In which case is diffraction effect more dominant — slit formed by 2 blades or slit formed by two fingers ?

(ii) Yellow light (λ = 6000 Å) illuminates a single slit of width 1×10^{-4} m. Calculate (i) the distance between two dark lines on either side of central maximum, in the diffraction pattern observed on a screen kept 1·5 m away from the slit, and (ii) the angular spread of the first minimum. *3*

OR

(b) (i) What will be the colour of the central bright fringe in Young's double slit experiment if the monochromatic source is replaced by a source of white light ? Give reason for your answer.

(ii) In Young's double slit experiment, the slits are separated by 0·3 mm and the screen is placed 1·5 m away from the slits. The distance between the central bright fringe and the sixth bright fringe is found to be 1·8 cm. Find the wavelength of light used in the experiment. *3*

30. (a) (i) An electron in a hydrogen atom jumps from second excited state to the first excited state. Name the spectral series in the spectrum of hydrogen atom to which the emitted radiation belongs.

(ii) Find the ratio of the wavelengths of the "most energetic spectral" lines in the Balmer series to that in Paschen series of Hydrogen spectrum. *3*

OR

(b) (i) An α-particle having kinetic energy K approaches a nucleus of atomic number Z. It gets close to the nucleus and then approaches a distance (d) and reverses its direction. Obtain an expression for the distance of closest approach (d) in terms of kinetic energy of the α-particle.

(ii) A proton and an alpha particle approach a target nucleus in head-on position, with equal velocities. Find the ratio of their distances of closest approach to the target nucleus. *3*

SECTION D

31. (a) (i) Draw a labelled ray diagram of an astronomical telescope to show the image formation of a distant object by it in normal adjustment. What are the main considerations required in selecting the objective and eyepiece lenses so that the telescope has large magnifying power and high resolution ?

(ii) A biconvex lens of focal length 20 cm is immersed in water, whose refractive index is $\frac{4}{3}$. Find the change, if any, in the nature and the focal length of the lens. Refractive index of the material of convex lens is $\frac{3}{2}$. *5*

OR

(b) (i) Draw a ray diagram showing refraction of light through a prism of angle A and obtain the relation between μ, A and the angle of minimum deviation δ_m.

(ii) An equiconvex lens of radius of curvature R and made of glass of refractive index μ is cut into two identical plano-convex lenses. Find the focal length of the plano-convex lenses. *5*

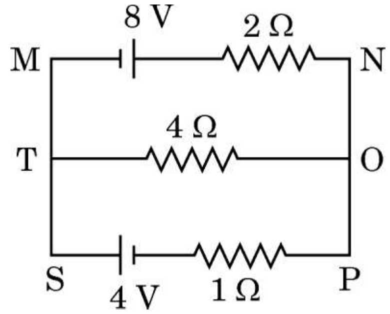
32. (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.

(ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1·5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be $\left(\frac{1}{2}\right)$ A and $\left(\frac{1}{3}\right)$ A respectively, calculate the internal resistance of each battery. *5*

OR

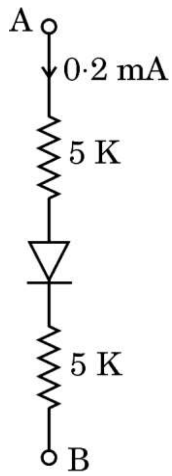
(b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.

(ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure. *5*



33. (a) (i) Draw the circuit diagram used to study I – V characteristics of a p-n junction diode in conducting mode. Mark on the graph the threshold voltage of the diode. Explain the significance of this voltage.

(ii) In the circuit shown in the figure, the forward voltage drop across the diode is 0·3 V. Find the voltage difference between A and B. *5*



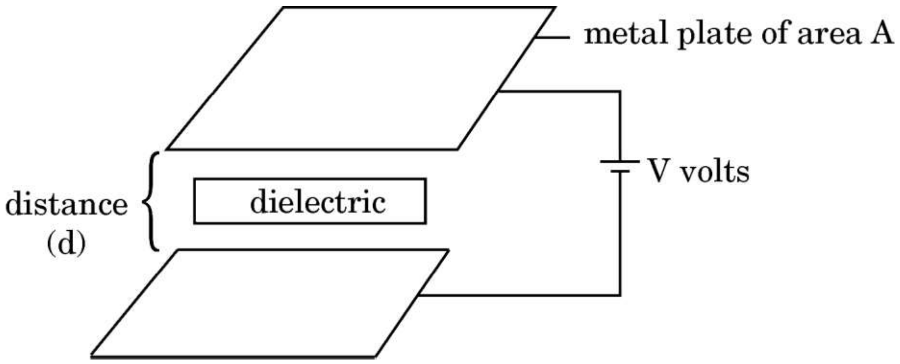
OR

(b) (i) Briefly describe the classification of solids into metals, insulators and semi-conductors on the basis of energy level diagrams.

(ii) In a silicon diode, the current increases from 10 mA to 20 mA when the voltage changes from 0·6 V to 0·7 V. Calculate the dynamic resistance of the diode. *5*

SECTION E

34.



A parallel plate capacitor is an arrangement of two identical metal plates kept parallel, a small distance apart. The capacitance of a capacitor depends on the size and separation of the two plates and also on the dielectric constant of the medium between the plates. Like resistors, capacitors can also be arranged in series or parallel or a combination of both. By virtue of electric field between the plates, charged capacitors store energy.

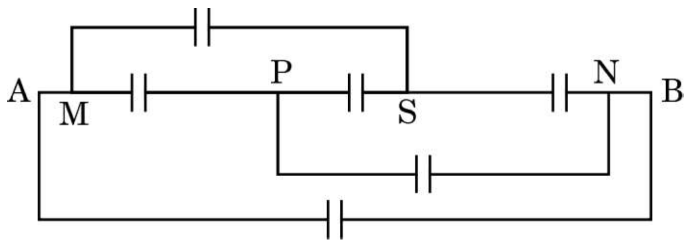
(a) The capacitance of a parallel plate capacitor increases from 10 μF to 80 μF on introducing a dielectric medium between the plates. Find the dielectric constant of the medium. *1*

(b) n capacitors, each of capacitance C, are connected in series. Find the equivalent capacitance of the combination. *1*

(c) A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected ? Justify your answer. *2*

OR

(c) Find the equivalent capacitance between points A and B, if capacitance of each capacitor is C. *2*



35. A prism is a solid transparent medium bounded by three rectangular faces with a triangular base and a top. A ray of light incident at angle i on one face of a prism suffers two refractions on passing through a prism. Hence it deviates through a certain angle δ from its original path. The angle of deviation becomes minimum ($\delta = \delta_m$) for a certain value of angle i. In such a condition, the refracted ray inside the prism becomes parallel to its base. An expression for refractive index μ of the material of the prism can be obtained in terms of angle A and angle δ_m.

(a) Show in a figure the variation of angle δ with angle of incidence i. *1*

(b) Show that for a prism of small angle A, the refractive index μ of its material can be written as $\mu = 1 + \frac{\delta_m}{A}$. *1*

(c) A ray of light passes through an equilateral prism such that both the angle of incidence and the angle of emergence are equal to the angle of prism A. Find the refractive index of the material of the prism, in terms of A. *2*

OR

(c) A ray of light passes through a prism of angle 75°, as shown in the figure. The refractive index of the material of the prism, with respect to its surrounding is $\sqrt{2}$. Find the angle of incidence i. *2*